

Malvern Madondo

CONTACT INFORMATION

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RESEARCH INTERESTS

Machine learning, reinforcement learning, dynamical systems, numerical optimization, and personalized educational technologies.

EDUCATION

Emory University, Atlanta, GA **August, 2019 - present**
Ph.D. Student, Computer Science and Informatics

The College of St. Scholastica, Duluth, MN **August, 2015 - May, 2019**
B.S., Computer Science/Information Systems and B.A., Mathematics

TECHNICAL SKILLS

- **Programming Languages:**
 - Most experience: Java, Python(incl. TensorFlow, Keras, PyTorch), Bash, Swift, SQL, HTML/CSS
 - Some Experience: MATLAB, PHP, Hack, JavaScript, React, C/C++
- **Version Control:** Git, Bitbucket, Mercurial
- **Operating Systems:** macOS, Linux, Windows
- **Methodologies:** Agile (SCRUM) and Waterfall

ACADEMIC EXPERIENCE

Emory University, Atlanta, GA

Graduate Student

August, 2019 - present

- **Reinforcement Learning for Mean Field Games:** Currently studying RL approaches to solving optimal control problems. Mentored by Lars Ruthotto.
- **Towards Student Mastery in Introductory Computer Science:** Striving towards understanding how student-authored question-answer pairs can make learning more effective in undergraduate courses. Mentored by Davide Fossati.

Instructor

August - December, 2020

I will be co-teaching CS 170, Introduction to Computer Science online. Will lead weekly recitation sessions and involve students in more hands-on learning activities.

Teaching Assistant - CS 171

January - May 2020

Assisted students with questions during lectures and office hours. Course emphasized the use and implementation of data structures, fundamental algorithms with introductory algorithm analysis, and object oriented design and programming with Java.

Teaching Assistant - CS 170

August - December 2019

Supported students taking the Intro to CS course by providing 1-on-1 instruction during lectures and office hours. Also graded assignments and quizzes. Topics included: fundamental computing concepts, general programming principles, the Unix Operating System, and the Java programming language.

The College of St. Scholastica, Duluth, MN

Math Teaching Assistant

August, 2018 - May 2019

Supported students taking Math courses such as Calculus and Linear Algebra, by providing 1-on-1 instruction and holding office hours.

Google IgniteCS Program - Lincoln Park Middle School, Duluth, MN

Mentor

January, 2018 - May 2018

Mentored Middle School students and taught weekly course in Computational Thinking as a foundation for Computer Science.

CodePath.Org - CSS, Duluth, MN

Technical Program Manager (iOS Dev Tutor)

September, 2017 - May 2019

Provided technical support for students during labs and assignments. Lead tutorials and manage in-class activities.

PUBLICATIONS

Madondo, M. and Gibbons, T. **Learning and Modeling Chaos Using LSTM Recurrent Neural Networks.** *Midwest Instruction and Computing Symposium (MICS 2018).*

http://micsymposium.org/mics2018/proceedings/MICS_2018_paper_26.pdf

Madondo, M., Gomez, DM., and Ciernia, N. **Hitchhiker's Guide to Computer Science for Social Good** *Midwest Instruction and Computing Symposium (MICS 2018).*

http://micsymposium.org/mics2018/proceedings/MICS_2018_paper_41.pdf.

POSTER
PRESENTATIONS

Madondo, M., Vang, J., and Gibbons, T. **Judging a whale by its tail: A Kaggle Humpback Whale Identification Challenge.** *MICS 2019*, NDSU, Fargo, North Dakota.

PROFESSIONAL
EXPERIENCE

Facebook - Crisis Response Team, Menlo Park, CA

Software Engineering Intern

May - August, 2018

- Worked with the Crisis Response/Safety Check team. Full-stack and back-end work in Hack, XHP, and JavaScript.
- Worked on bringing Community Help, which allows users to find and offer help during a crisis, to Facebook's mobile site.
- Built a Mark Safe QP on WWW platform that resulted in a 25% increase in primary clicks, 43% increase in QP dismissals and 50% increase in 'Does Not Apply' clicks.

Facebook - FBU, Menlo Park, CA

Software Engineering Intern

June - August, 2017

- Participated in the Facebook University for Engineering program – iOS Development track.
- Developed from scratch an iOS app that integrated location-based facts, Tinder-style fact sharing, localization, image recognition (utilizing IBM Watson API) and Wikipedia descriptions.

The College of St. Scholastica - Marketing Department, Duluth, MN

Web Assistant

October, 2016 - May, 2019

- Supported the web and marketing goals of the Interactive Services Team through web content design, creation, and editing, while following branding guidelines and best practices.
- Prepared site pages and content to meet accessibility guidelines.

State of Minnesota - Department of Human Services, St. Paul, MN

Urban Scholar and Management Analyst

May - August, 2016

- Assisted the Member Provider Services communications team in the development and implementation of a process to archive Provider Manual web page revisions and index web page documents.
- Implemented the Minnesota State Government's Plain Language initiative in web page revisions

- to promote accessibility.
- Participated in the Urban Scholars Leadership Institute. Awarded Certificate of Commendation.

The College of St. Scholastica - IT Help Desk, Duluth, MN

Team Lead and Tech Assistant

September, 2015 - October, 2019

- Co-led and supervised a team of 15 Tech Assistants offering technical support, troubleshooting and diagnosing network and technical issues.
- Demonstrated advanced customer service skills while interacting and communicating with peers, faculty, and staff from diverse backgrounds.

**HONORS AND
AWARDS**

2020 Science ATL Communication Fellow

2020 Computing Research Association ([CRA-WP](#)) Grad Cohort for Underrepresented Minorities and Persons with Disabilities (URMD) scholarship recipient

2019 The College of St. Scholastica: graduated summa cum laude

2019 Webster Scholar (Honors program), The College of St. Scholastica

2018 Minnesota High Tech Association Foundation (MHTF) scholarship recipient

2018 Affirm scholarship recipient for the Richard Tapia Conference

2018 Quip Diversity Scholarship runner-up

2018 Student Employee of the Year Award, The College of St. Scholastica

2017 Code2040 Tech Trek fellow

2017 Minnesota Private Colleges, Phillips Scholarship recipient

2016 Minneapolis Urban Scholars Leadership Award

Philanthropic Ventures Foundation Grace Scholarship recipient (2015-19)

**LEADERSHIP AND
OTHER ACTIVITIES**

NORTHFORCE — Mentor Connection Program, Duluth, MN

Student Ambassador

October, 2016 - May, 2019

Connect and network with local professionals to learn about career options and gain real world insights.

Marketing Department — The College of St. Scholastica, Duluth, MN

Blogger

September, 2015 - May, 2019

Maintained department's social media presence to engage current and prospective students. Provided insights into student life while sharing experiences as a student.

WellU Program — The College of St. Scholastica, Duluth, MN

Storm's Advocate

August, 2016 - May, 2018

Present educational programs and table at events promoting a lifestyle of health and wellness to college community.

DIGI-KEY Electronics — Programming Competition, Thief River Falls, MN

Competitor

October, 2017

Mentored by professor Kris Glesener and in a team of four, placed fifth place overall. Won first place in two "blender" sessions.

Office of International Programs — The College of St. Scholastica, Duluth, MN

Student Ambassador

August — September, 2016

Assisted in the organization and implementation of the summer orientation program for incoming international students.

Youth Theology Institute — The College of St. Scholastica, Duluth, MN

Peer Leader

July, 2016

Mentored participants and led opportunities for community building and small group discussions.

Midwest Instruction & Computing Symposium — Robotics Competition

Competitor

April, 2016 and April, 2017

Designed and programmed an autonomous robot that could trace a path and play a 3-hole miniature golf course, with the help of two other students and advice from a faculty.

Mathematical Association of America, NCS — Team Competition

Competitor

November, 2015-2018

Participated in the annual math competition organized by the [Mathematical Association of America, North Central Section](#). Problem level ranged from moderately easy to difficult Putnam level Math problems.

LANGUAGES

English (Full Proficiency)

Shona (Full Proficiency)

French (Elementary Proficiency)